During the one-second refresh interval, a new segment is created to hold however many documents are collected in the in-memory buffer. The heap memory of the shard (Lucene instance) dictates the number of documents it can hold in an in-memory buffer before flushing them out to the file store. For all practical purposes, Lucene treats the segments as immutable resources. That is, once a segment is created with available documents from the buffer, no new documents will make their way into this existing segment. Instead, they're moved into a new segment. Similarly, deletes are not performed physically on documents in segments, but documents are marked for removal later. Lucene employs this strategy to provide high performance and throughput.

5.1.3 Customizing the refresh process

Indexed documents live in memory until the refresh cycle kicks in. This means we have uncommitted (non-durable) documents until the next refresh cycle. A server failure can trigger a data loss. The risk of data loss increases as the refresh period increases (a one-second refresh cycle time has a lower risk of losing data than a one-minute refresh cycle time). On the other hand, reducing the refresh time leads to more I/O operations, which can cause performance bottlenecks—so we must find the optimal refresh strategy for our organization.

Elasticsearch is an *eventually consistent* application—meaning documents are eventually written to the durable store. Documents are moved into the filesystem as segments during the refresh process and thus are available to search. The refresh process is expensive, especially if the engine is hammered with a large number of indexing requests.

CONFIGURING THE REFRESH CYCLE

The good news is that we can configure this refresh setup. We can reset the time interval from the default 1 second to, say, 60 seconds by tweaking the settings on an index level using the `_settings` endpoint.

Listing 5.7 Setting a custom refresh interval

```
PUT movies/_settings
{
  "index":{
    "refresh_interval":"60s"
  }
}
```

This is a dynamic setting, which means we can change the refresh setting on a live index at any time. To switch off the refresh operation completely, set the value to `-1`. The in-memory buffer will accumulate incoming documents if the refresh operation is off. The use case for this scenario may be that we are migrating many documents from a database into Elasticsearch, and we don't want data to be searchable until the